Decision-Calibrated Paired Evaluation with Sequentially Acquired Human Judgments

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Abstract

Pairwise language-model evaluations increasingly use inexpensive model judges, but deployment decisions concern a specified human population and must account for ties, prompt variation, correlated judge errors, and adaptively chosen human labels. We formulate decision-calibrated paired evaluation around the finite-frame estimand "human win probability minus loss probability" under fixed item, prompt, response-generation, rater, and display-order distributions. A tie-aware hierarchical jury model drives one-step value-of-information acquisition, while an explicit random-exploration mixture and logged draw probabilities preserve design-based identification. A prequential augmented Hansen–Hurwitz estimator is conditionally unbiased under stated response-stability, positivity, and boundedness conditions and supports fixed-budget martingale inference even when the jury model is misspecified. This methods-first draft specifies simulation, human-audit, and temporal-shift tests; efficiency and decision-quality claims remain pending empirical execution.

1. Introduction

Model comparison is often reduced to a leaderboard score, although the actual decision may be whether system A should replace system B for a particular population of requests. Human pairwise evaluation directly targets preference and underlies Chatbot Arena (Chiang et al., 2024), but comprehensive human labeling is expensive. Model-based judges scale cheaply (Zheng et al., 2023) yet exhibit position, style, verbosity, and self-preference effects (Panickssery et al., 2024; Ye et al., 2025). A jury of correlated judges cannot be treated as an equally large set of independent humans.

Recent work estimates judge-specific bias and uncertainty (Dubois et al., 2026), shared confounding (Zhao et al., 2026), jury discrimination and internal consistency (Qian et al., 2026), and uncertainty with iterative comparison selection in comparative evaluation (Fathullah & Gales, 2025). Human-calibrated estimators correct noisy model judgments (Lee et al., 2026; Chen et al., 2026). Efficient inference for a static, independently selected calibration set is now well understood enough that "apply AIPW to judge labels" is not a new contribution.

The specific issue studied here is the combination of a tie-aware crossed human-preference frame, decision-loss acquisition, and sequential inference. Once human cells are chosen because previous outcomes made them decision-relevant, a final inclusion probability cannot generally be reconstructed by multiplying probabilities along the realized path. Ordinary post-hoc cross-fitting can leak future selection information, while pure value-of-information acquisition can assign zero probability to target cells.

We ask: *can adaptive human labeling reduce the cost of deciding whether A is materially better than B while retaining valid inference for an explicit human-preference estimand?* We make four scoped contributions:

- We define a tie-aware finite evaluation frame that fixes item, prompt, response coupling, rater, and display-order distributions before labeling.
- We use a crossed jury model only for prediction and decision-targeted acquisition, not as the definition of quality.
- We combine value-of-information acquisition with randomized exploration and a draw-level sequential augmented Hansen–Hurwitz estimator. We state its response-and-design assumptions and fixed-budget asymptotic conditions.
- We lay out planned simulations, a sealed target-random human audit, shift tests, and ablations that separate interval validity from label efficiency; executable DGPs remain a preregistration blocker.

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Evidence status. The formal design claims are established below. Whether adaptive acquisition saves labels, whether the working model predicts humans, and whether the chosen human panel represents deployment stakeholders remain empirical questions; no such result is reported in this draft.

2. Related Work

Measurement variation and judge bias. Multi-prompt evaluation models performance distributions across prompt forms (Polo et al., 2024). Human studies can be assessed by the probability that a repeat study preserves a ranking (Riley et al., 2024). Model judges introduce additional structure: self-recognition can produce self-preference (Panickssery et al., 2024), controlled perturbations reveal multiple biases (Ye et al., 2025), and modern hierarchical methods estimate grader effects, confounding, or discrimination (Dubois et al., 2026; Zhao et al., 2026; Qian et al., 2026). TrustJudge formalizes score–comparison and tie-transitivity inconsistencies and aggregates bidirectional likelihoods to reduce them (Wang et al., 2026), while distribution-calibrated inference-time compute fits a Bradley–Terry–Davidson model to repeated thinking-judge vote counts and aligns aggregation with a target loss (Dadkhahi et al., 2026). Our jury model borrows this measurement perspective. Its novelty is not a new bias taxonomy or latent quality model, and calibrated judge vote counts remain predictive features rather than the target human estimand.

Human-calibrated inference. Binary sensitivity/specificity correction with propagated calibration uncertainty and pilot-based allocation across true-label classes is studied by Lee et al. (2026). Chen et al. (2026) unify measurement-error and prediction-powered estimators for mean human outcomes, including pairwise win rates, under independently selected calibration/test designs. Those estimators do not by themselves justify reusing an outcome-adaptively acquired calibration sample. Classical unequal-probability with-replacement estimation (Hansen & Hurwitz, 1943) and inference for adaptively collected data (Hadad et al., 2021) supply the statistical foundation. Prediction-Powered Active Testing (PPAT) combines active-testing weights with a prediction control variate (Ashouritaklimi et al., 2026); Active Testing supplies the earlier adaptive risk estimation framework (Kossen et al., 2021), while control-variate evaluation combines synthetic and human feedback for unbiased head-to-head win rates (Zhou et al., 2025). Our distinction is decision-loss acquisition for three-way human comparisons over a crossed evaluation frame, not adaptive residualization or its central limit theorem by itself. DMLRank gives nonparametric, debiased inference for tie-aware preference rankings and derives budget-aware collection policies (Frauen et al., 2026). Its target is a multi-model ranking functional; ours is one frozen replacement decision for an explicitly weighted item–response–rater–order frame, with outcome-adaptive draw probabilities logged prequentially. Celeus combines uncertainty-guided sampling and surrogate predictions in conditionally unbiased signals and builds anytime-valid e-process intervals for a scalar evaluation score (Zhou et al., 2026). SERPANT uses pairwise e-processes for online model ranking with family-wise error control under continuous monitoring (Gu et al., 2026). Both permit data-dependent stopping; our martingale central-limit interval does not and therefore fixes T. An e-process version of Equation (8) is a natural extension, but it would change the terminal interval and is not claimed here.

Adaptive and dynamic evaluation. Active acquisition can exploit historical dependence among benchmark items (Li et al., 2025b), while ProEval uses surrogate models for performance estimation and failure discovery (Huang et al., 2026). Such compression can fail under source–target model shift (Zhang et al., 2025). Dynamic benchmarks reduce static contamination through generated or refreshed tasks (Zhu et al., 2024; White et al., 2025); contamination tests can detect some forms of memorized ordering (Oren et al., 2024). Freshness, however, changes the evaluation distribution and complicates longitudinal claims. ATLAS uses item-response information to adaptively estimate a scalar model ability from static benchmark items (Li et al., 2025a). It is a relevant efficiency comparator, but its calibrated-item-bank target differs from our crossed human pairwise mean and four-action replacement decision. Finally, non-transitive judge preferences can make a scalar rank baseline-dependent (Xu et al., 2025). We therefore target one prespecified pairwise replacement decision, not a universal leaderboard. SCOPE conformally calibrates abstention for pairwise model judgments under exchangeability (Badshah et al., 2026); its selective-judge risk is distinct from estimating a weighted human-panel preference with adaptive labels. Collaborative adaptive labeling also combines imperfect labels with selective expert escalation and dependence-aware estimation (Ruan et al., 2026). FairJudge instead trains the judge itself to reduce selected biases (Yang et al., 2026). Our model jury remains fallible: identification comes from randomized human acquisition, and the claimed novelty is limited to the tie-aware crossed decision frame and decision-loss policy. AURA adaptively prioritizes uncertain pairwise judge decisions for human verification (Zhang et al., 2026). It targets refinement of latent judge trust; our target is instead a prespecified weighted human-panel mean, with the acquisition model allowed to be wrong. For reward learning, near-tie pairs are the most informative queries, and instance-dependent designs exploit this to achieve nearly optimal label complexity (Zhao & Jun, 2026). Our acquisition targets a different quantity. EVI rewards a label only if it can change the four-way action on Δ_F, and logged probabili-

ties keep the design-based estimator valid even when the working model is wrong.

3. Evaluation Goal and Desiderata

At evaluation time t_0, freeze models A, B, their decoding policies, the rubric, and a finite frame of N root items. Item i has target weight $w_i \geq 0$, $\sum_i w_i = 1$. Cell $c \in \mathcal{C}_i$ identifies a prompt variant and paired response realizations with prespecified conditional weight $a_{c|i}$. Independent response draws, shared random seeds, and deterministic decoding define different estimands because preference is nonlinear; the coupling is therefore fixed explicitly.

Before acquisition, enroll a finite rater panel $\mathcal{H} = \{1, \ldots, H\}$ with nonnegative target weights ρ_h summing to one. Inference therefore targets this panel; extrapolation to other people is a separate superpopulation claim. For display order $o \in \{-1, +1\}$, define the outcome of one standardized rating draw

$$Y_{icho} = \begin{cases} +1, & A \text{ preferred}, \\ 0, & \text{tie}, \\ -1, & B \text{ preferred}. \end{cases} \tag{1}$$

and $\mu_{icho} = \mathbb{E}[Y_{icho} \mid i, c, h, o]$, where the expectation covers within-rater response variation under the fixed protocol. The finite-frame target is

$$\Delta_F = \sum_{i=1}^{N} w_i \sum_{c \in \mathcal{C}_i} a_{c|i} \sum_{h \in \mathcal{H}} \rho_h \frac{1}{2} \sum_{o \in \{-1,+1\}} \mu_{icho}. \tag{2}$$

Equivalently, define a target unit $x = (i, c, h, o) \in \mathcal{X}$ with mass

$$\pi(x) = w_i a_{c|i} \rho_h / 2, \qquad \sum_{x \in \mathcal{X}} \pi(x) = 1. \tag{3}$$

Then $\Delta_F = P_\pi(A \text{ wins}) - P_\pi(B \text{ wins})$ and the half-tie win rate is $(1 + \Delta_F)/2$. We separately target

$$\tau_F = \sum_{x \in \mathcal{X}} \pi(x) P(Y_x = 0 \mid x) \tag{4}$$

because the same Δ_F can describe decisive disagreement or widespread ties.

A practical margin $\delta > 0$ and loss matrix are pre-specified. The action set is "A materially better" if $\Delta_F > \delta$, "practically equivalent" if $|\Delta_F| \leq \delta$, "B materially better" if $\Delta_F < -\delta$, and "not established" when data do not support a decisive action. This is an operational human-preference construct, not latent universal quality.

4. Evaluation Method

The protocol first collects inexpensive labels from a family-diverse model jury on all cells, then an IID target-random human pilot from π. Before each subsequent batch, a working model is trained only on prior batches and used to compute decision value. Human acquisition mixes this adaptive policy with the target distribution. A sealed IID target-random human audit is never used for fitting or acquisition. The implementation refits once per predeclared batch. It approximates each hypothetical EVI update by one frozen Laplace/Newton step, forms one p_b for the next batch, and logs that probability for every draw. The working Bayes-risk action used inside EVI is distinct from the terminal confidence-set action. Conditional on the batch-start history, draws are IID with replacement from this frozen law and the predictor is fixed; draw-level Equation (8) remains valid because both are predictable.

4.1. Data and Task Construction

The frame is a probability sample from a timestamped deployment log or a weighted benchmark with a clearly narrower finite-frame interpretation. Sampling excludes personal data, unsupported licenses, and items requiring professional expertise absent from the rater pool. Root-item provenance, inclusion weight, domain, language, safety category, near-duplicate cluster, prompt generator, response seeds, model revisions, tokenizer, and decoding parameters are versioned before labels are revealed. Each public root also records its pinned repository revision, export-byte hash, stable upstream identifier, canonical raw-row hash, and versioned normalized-content hash. Normalized duplicate clusters are formed before frame sampling and assigned as units; any separately admitted pilot-fitting or sealed source is a distinct registered role, and cross-role clusters are quarantined. Paraphrase and template leakage remain separate blocking checks, since hash inequality is not evidence of semantic independence. A hash-bound preflight additionally rejects named PII/secret patterns and retained character-5-gram near-duplicates; it is a lexical heuristic, not a substitute for reviewed multilingual, paraphrase, privacy, or benchmark-leakage audits. Above the exhaustive limit, only a reviewed exact engine whose hash-bound shards cover every cross-record field segment can lift the lexical blocker. The committed engine is unfrozen and indexes record pairs only, so it cannot clear a real multi-field frame.

Each prompt variant is validated as meaning-preserving by a separate process; the target weights over variants are fixed, not chosen after seeing which model benefits. Model identifiers are hidden (although style may reveal identity), AB/BA display is randomized, and humans receive a real tie option with a fixed definition. "Equally good" and "both unacceptable" are recorded separately even if the primary analysis maps both to zero. Independent ratings define Equation (2); majority vote and adjudication are secondary, different estimands.

Rater and order assignments follow the target masses in Equation (3), or their logged adaptive counterparts. Availability is resolved before content is shown. Every assignment attempt is logged; a skip after viewing content is a separate outcome, never silently reassigned. The formal estimator below applies only to complete ratings. For nonresponse we report worst-case bounds and a separately labeled inverse-response analysis under a stated missing-at-random model; outcome-dependent skipping is not repaired by weighting alone.

4.2. Scoring and Aggregation

At round t, sample one complete target unit with replacement,

$$X_t = (I_t, C_t, H_t, O_t) \sim p_t(\cdot \mid \mathcal{F}_{t-1}), \tag{5}$$

where $\mathcal{F}_{t-1}$ contains every prior assignment, sampling probability, jury feature, outcome, and model random seed. This joint law makes rater and order assignment explicit. If operational constraints require a nested assignment, the product of logged item, cell, rater, and order probabilities must equal $p_t(X_t)$. Write $Y_t^\star$ for the standardized rating that would be recorded under a completed assignment. The design claim additionally requires stability of its full response law:

$$P(Y_t^\star = y \mid \mathcal{F}_{t-1}, X_t = x) = p_y(x), \quad y \in \{-1, 0, 1\}, \tag{6}$$

where $\mu_x = \sum_y y p_y(x)$. Learning, fatigue, memory, or rubric drift can violate Equation (6). Randomized task order, training before study labels, and temporal diagnostics reduce but do not prove this assumption. The protocol randomizes attempt positions, interleaves hidden audit attempts, and retains repeated roots rather than resampling them. A sensitivity analysis assumes the weaker uniform bound $|\mathbb{E}[Y_t \mid \mathcal{F}_{t-1}, X_t = x] - \mu_x| \le \gamma$; then the design estimator's absolute drift bias is at most γ, so both interval endpoints are expanded by the frozen $\gamma \in \{0, .02, .05\}$.

Let $m_{t-1}(x) \in [-1, 1]$ be a bounded prediction of Y_x trained only on prior rounds and frozen before draw t, and define

$$\bar{m}_{t-1} = \sum_{x \in \mathcal{X}} \pi(x) m_{t-1}(x). \tag{7}$$

The draw-level sequential augmented Hansen–Hurwitz contribution is

$$\psi_t = \bar{m}_{t-1} + \frac{\pi(X_t)}{p_t(X_t)} \{Y_{X_t} - m_{t-1}(X_t)\}, \tag{8}$$

with final estimator

$$\widehat{\Delta}_T = \frac{1}{T} \sum_{t=1}^{T} \psi_t. \tag{9}$$

Repeated draws are retained. p_t is the conditional draw probability, not the probability that a unit is ever included; substituting a realized-path inclusion probability is generally invalid under outcome adaptation. Replacing Y_x in Equation (8) by $\mathbf{1}\{Y_x = 0\}$ gives the tie-rate estimator, with its covariance with $\widehat{\Delta}_T$ reported.

The fixed T counts assignment attempts, never replacement completions. Let R_t indicate observation and store $Y_t^{\text{obs}} = R_t Y_t^\star$, where $Y_t^\star \in [-1, 1]$; let $\psi_t^\star$ denote Equation (8) with this complete outcome. The primary report replaces a missing outcome pointwise by -1 and $+1$ in

$$\psi_t^{\pm} = \bar{m}_{t-1} + \frac{\pi(X_t)}{p_t(X_t)} \left[R_t \{Y_t^{\text{obs}} - m_{t-1}(X_t)\} + (1 - R_t)\{\pm 1 - m_{t-1}(X_t)\} \right]. \tag{10}$$

Thus $\psi_t^- \le \psi_t^\star \le \psi_t^+$ for every assignment, without a missing-at-random assumption. Exact interval construction appears in Section 5.2. A separate analysis assumes $P(R_t = 1 \mid \mathcal{F}_{t-1}, X_t, Y_t^\star) = e_t(X_t) \ge e_{\min} > 0$ and uses the predictable contribution

$$\psi_t^{\text{MAR}} = \bar{m}_{t-1} + \frac{\pi(X_t)}{p_t(X_t)} \frac{R_t}{e_t(X_t)} \{Y_t^{\text{obs}} - m_{t-1}(X_t)\}.$$

It is secondary because an estimated response model does not make outcome-dependent skipping ignorable.

4.3. Human or Model-Based Judging

The jury model is a prediction and diagnostic instrument, not the source of identification. A covariate-dependent generalization of the Davidson tie model (Davidson, 1970) uses

$$D_{jico} = e^{\eta_{jico}/2} + e^{-\eta_{jico}/2} + e^{\kappa_{jico}}, \tag{11}$$

$$P_j(Y = 1) = e^{\eta_{jico}/2} / D_{jico}, \tag{12}$$

$$P_j(Y = -1) = e^{-\eta_{jico}/2} / D_{jico}, \tag{13}$$

$$P_j(Y = 0) = e^{\kappa_{jico}} / D_{jico}. \tag{14}$$

One explicit working predictor is

$$\eta_{jico} = \alpha_j + d_j(\beta + b_i + b_{iv}) + \gamma_j^O o + \gamma_j^L \ell_{ic} + \gamma_j^S s_{jic} + u_{j,d(i)}, \tag{15}$$

$$\kappa_{jico} = \kappa_j + t_i + t_{iv} + z_{ic}^\top \xi_j, \tag{16}$$

with $d_j > 0$ and partially pooled effects. Identifiability constraints set weighted human-panel offset to zero and weighted log-discrimination to zero, center item and variant effects, and require a connected model-judge–cell overlap graph with both orders and estimable covariate contrasts. The public pilot has one checkpoint per jury family, so it does not estimate a separate family-variance component. Human-rater effects are shrinkage-only working predictions under a proper prior; the 200-attempt pilot does not support a pilot-era rater-heterogeneity claim or require every rater to

appear. Conditional independence is only a working assumption. The two recorded tie subtypes are collapsed to zero before this three-category model and analyzed separately.

The human scale is anchored by the frozen weighted panel rather than assuming jury consistency equals validity. Length and family coefficients remain descriptive unless randomized interventions identify them: removing a length association can remove genuine quality as well as bias.

Decision-targeted acquisition. The protocol fixes the normalized loss matrix below, with rows (A, B, E, NE) and true states (A, B, E):

$$L = \begin{pmatrix} 0 & 4 & 2 \\ 4 & 0 & 2 \\ 2 & 2 & 0 \\ 1 & 1 & 1 \end{pmatrix}. \tag{17}$$

Stakeholder-specific alternatives are sensitivity analyses, not post-hoc replacements. Let ϑ collect the finite working-model coefficients. At batch start b, freeze the Laplace posterior $q_b(\vartheta) = \mathcal{N}(\widehat{\vartheta}_b, H_b^{-1})$, the deterministic frame functional $\Delta(\vartheta)$, and

$$Q_b(S = s) = \int \mathbf{1}\{S(\Delta(\vartheta)) = s\} q_b(\vartheta) d\vartheta, \tag{18}$$

$$\widehat{p}_b(y \mid x) = \int p_\vartheta(y \mid x) q_b(\vartheta) d\vartheta. \tag{19}$$

Both integrals use frozen common Monte Carlo draws. Define Bayes risk

$$\widetilde{R}(Q) = \min_{d \in \{A,B,E,\mathrm{NE}\}} \sum_s L_{ds} Q(S = s), \tag{20}$$

breaking action ties in the fixed order NE, E, A, B. Let $q_b^{x,y}$ be obtained from q_b by one predeclared damped Newton update of the mode and observed Hessian using $\log p_\vartheta(y \mid x)$, with frozen prior, ridge, damping, and no-line-search rule. Then $U(Q_b; x, y)$ is the state distribution induced by $q_b^{x,y}$ through the same $\Delta(\vartheta)$ and common draws. Neither the update nor its hyperparameters depend on p_b. The update evaluates gradient and Hessian at $\widehat{\vartheta}_b$, uses the frozen ridge/damping formula in the preregistration, and factorizes with a fixed jitter ladder. Standard-normal draws use seed 2026090322. If every registered Cholesky attempt fails, that cell receives EVI zero and the failure is retained rather than silently changing the approximation. For target unit x with annotation cost k_x, define

$$\mathrm{EVI}_b(x) = \widetilde{R}(Q_b) - \sum_{y \in \{-1,0,1\}} \widehat{p}_b(y \mid x) \widetilde{R}\{U(Q_b; x, y)\}. \tag{21}$$

At each adaptive batch start, draw a frozen $K = 256$ candidate multiset $\mathcal{C}_b$ IID with replacement from π, before computing EVI. Let $n_b(x)$ be its multiplicity. Using predeclared temperature λ_E, define

$$g_b(x) = \lambda_E \, \mathrm{EVI}_b(x)/k_x, \tag{22}$$

$$r_b(x) = \frac{n_b(x) \exp\{g_b(x)\}}{\sum_{x'} n_b(x') \exp\{g_b(x')\}}, \tag{23}$$

$$p_b(x) = \epsilon \pi(x) + (1 - \epsilon) r_b(x), \qquad \epsilon > 0. \tag{24}$$

Here $n_b(x) = 0$ sets $r_b(x) = 0$; duplicates are retained through their multiplicity. A domain-separated seed schedule and log-sum-exp implementation are frozen. Across 40 adaptive batches this caps computation at 30,720 hypothetical Newton updates and 125,829,120 posterior-draw evaluations. All EVI values are computed before p_b, after which Q_b, p_b, m_b, and $\bar{m}_b$ are frozen for the complete batch; each draw-level p_t equals that p_b. Before the batch's first draw, one commitment hash binds its candidate multiset, EVI values, annotation costs, complete prediction maps, and the list of earlier slots used for fitting. Every logged p_t must then be recomputable from that commitment alone. This removes any dependence of EVI on its own acquisition probability and gives $p_t(x) \geq \epsilon \pi(x)$ and $\pi(x)/p_t(x) \leq 1/\epsilon$. Exact EVI is myopically optimal only under a correct posterior, exact update, equal costs, and argmax acquisition. The implemented finite-temperature candidate-mixture policy has no such optimality claim, finite-horizon guarantee, or general regret bound. It scores approximate Bayesian risk reduction, whereas the terminal action uses a confidence set; their alignment is a required simulation diagnostic, not an identity. Exact preposterior EVI is nonnegative, but the one-step Laplace update is not a martingale for Q_b, so the implemented score can be negative. The design also caps how much EVI can matter: with costs $k_x = 1$, the softmax tilt between any two candidates is $\exp\{\lambda_E(\mathrm{EVI}_{\max} - \mathrm{EVI}_{\min})\}$. On the generated golden law of Appendix A, the registered $\lambda_E = 2$ gives a tilt of at most 1.11 after 20 fit outcomes and at most 1.008 from 40 to 200 outcomes. One-step EVI is negative for every unit at 40 and 120 fit outcomes. These are properties of a constructed three-parameter law, not pilot estimates. They show that the simulation study must report whether the adaptive arm differs materially from target-random sampling with candidate multiplicities before any efficiency gain is attributed to EVI.

5. Validation

Proposition 5.1 (Sequential unbiasedness)**.** *Assume the finite frame and fixed budget T are prespecified; Equation (6) holds; outcomes are complete; p_t and m_{t-1} are measurable before outcome t; sampling is with replacement from the logged p_t; and Equation (24) holds. Then*

$$\mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}] = \Delta_F, \qquad \mathbb{E}[\widehat{\Delta}_T] = \Delta_F, \tag{25}$$

for any misspecified working jury model or outcome predictor.

The proof follows by conditioning on X_t and appears in Appendix A. Forward or prequential fitting is essential: each prediction and sampling probability is frozen before its outcome. Ordinary random K-fold fitting after adaptive collection can leak information through future selections.

Theorem 5.2 (Fixed-budget inference)**.** *Under the assumptions of Proposition 5.1, let $D_{Tt} = \psi_{Tt} - \Delta_F$. Suppose*

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[D_{Tt}^2 \mid \mathcal{F}_{t-1}] \xrightarrow{p} V \in (0,\infty) \tag{26}$$

for deterministic V, and for every $\eta > 0$ the martingale Lindeberg quantity

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}[D_{Tt}^2\mathbf{1}\{|D_{Tt}| > \eta\sqrt{T}\} \mid \mathcal{F}_{t-1}] \xrightarrow{p} 0. \tag{27}$$

Then

$$\sqrt{T}(\widehat{\Delta}_T - \Delta_F) \xrightarrow{d} \mathcal{N}(0, V). \tag{28}$$

This is the standard martingale central-limit route (Hall & Heyde, 1980), specialized to predictable adaptive sampling. Because $Y, m \in [-1,1]$ and $\pi/p \le 1/\epsilon$, increments are bounded when ϵ is fixed. This boundedness gives the required squared-increment martingale law of large numbers, so

$$\widehat{V}_T = \frac{1}{T-1}\sum_{t=1}^{T}(\psi_t - \widehat{\Delta}_T)^2, \qquad \widehat{\mathrm{se}} = \sqrt{\widehat{V}_T/T} \tag{29}$$

gives studentized fixed-budget inference. This is pointwise asymptotic, not exact finite-sample, coverage. The protocol fixes T before the first human outcome and forbids threshold-based optional stopping.

5.1. Contamination and Gaming Resistance

Public static items are checked for exact, normalized, paraphrase, and template-level overlap with model-accessible corpora where those corpora are available. Ordering-based tests provide evidence for only particular forms of memorization (Oren et al., 2024); absence of a signal is not proof of cleanliness. A fresh private wave and generated variants reduce exposure but create a new population, so longitudinal analyses retain bridge items and separate within-wave model difference from between-wave drift.

Judge prompts and identities are hidden from evaluated systems. Position is randomized, response identifiers are stripped, and controlled style/length transformations probe judge sensitivity. Model outputs containing instructions to the evaluator are separately flagged. The human-only sealed audit remains the final check against jury manipulation.

5.2. Statistical Protocol

When every assignment is complete, the primary two-sided interval uses the frozen $z_{0.975} = 1.959963984540054$:

$$[L_T^{\mathrm{cmp}}, U_T^{\mathrm{cmp}}] = \left[\widehat{\Delta}_T \pm z_{0.975}\sqrt{\widehat{V}_T/T}\right] \cap [-1,1]. \tag{30}$$

If any $R_t = 0$, define $B_\epsilon = 1 + 2/\epsilon$, $c_\alpha = B_\epsilon\sqrt{2\log(2/\alpha)/T}$, and use instead

$$\begin{aligned} L_T^{\mathrm{miss}} &= \max\left\{-1, \frac{1}{T}\sum_t \psi_t^- - c_\alpha\right\}, \\ U_T^{\mathrm{miss}} &= \min\left\{1, \frac{1}{T}\sum_t \psi_t^+ + c_\alpha\right\}. \end{aligned} \tag{31}$$

Each endpoint contribution lies in $[-B_\epsilon, B_\epsilon]$ and its predictable mean bounds Δ_F; two one-sided martingale–Hoeffding bounds give coverage at least $1-\alpha$ without a response-mechanism assumption. Write $[L_T, U_T]$ for the applicable interval. It is widened by γ in the response-drift sensitivity analysis. The tie-rate analogue replaces a missing indicator by 0 and 1. Its contributions lie in an interval of length B_ϵ, so its radius is $B_\epsilon\sqrt{\log(2/\alpha)/(2T)}$, clipped to $[0,1]$. This guarantee is intentionally severe: at the pilot values $\epsilon = 0.2$, $T = 1000$, and $\alpha = 0.05$, $c_\alpha = 0.9448$. Consequently even one missing outcome will generally force NE; simulation must report this degeneracy rather than presenting the bound as label-efficient. The terminal action used with Equation (17) and in the final report is

$$\widehat{a} = \begin{cases} A, & L_T > \delta, \\ B, & U_T < -\delta, \\ E, & [L_T, U_T] \subseteq [-\delta, \delta], \\ \mathrm{NE}, & \text{otherwise.} \end{cases} \tag{32}$$

For complete outcomes this wrong-action probability is at most $\alpha + o(1)$; with Equation (31) it is at most α under response stability, for this single comparison. This does not bound abstention, expected stakeholder loss, or multi-model ranking error.

Theorem 5.2 is conditional on the full realized frame, including the enrolled rater panel and fixed response pairs. For a separate deployment-population target Δ_{pop},

$$\widehat{\Delta}_T - \Delta_{\mathrm{pop}} = (\widehat{\Delta}_T - \Delta_F) + (\Delta_F - \Delta_{\mathrm{pop}}). \tag{33}$$

The martingale variance covers only the first term. The second requires the actual first-stage item and rater recruitment design and its replicate weights or a justified superpopulation model; prompt variants cannot be bootstrapped as independent root items.

6. Empirical Study

Simulation precedes human collection. We vary Δ_F at $\{-\delta, 0, \delta\}$ and local alternatives around both decision boundaries. A full Cartesian crossing of all stressors is neither computationally credible nor the registered design. Instead, a frozen 16-run two-level fraction varies tie prevalence, item and judge-family correlation, repeated-rater variation, nonresponse, rare-stratum mass, exploration, and temporal drift. Its eight main-effect columns are balanced and orthogonal, with no main effect aliased with a two-factor interaction; two-factor interactions remain aliased. We cross those rows with three boundary effects and three working-model regimes, then add center-condition boundary and local-alternative blocks, for 189 conditions total.

For each condition, 2,000 Monte Carlo trials compare target-random human sampling, static independent calibration, Active Testing, control-variate evaluation, PPAT-style residual acquisition, variance-optimal residual sampling, pure uncertainty sampling, and EVI. Within a condition and trial, policies share the latent frame and complete potential-outcome stream but use policy-keyed randomization. Each acquisition policy is crossed with the same valid Equation (9); augmentation is then ablated at fixed policy. Exact DGP and policy implementations remain hash-blocked. Until faithful native implementations are frozen, Active Testing, control-variate, and PPAT names refer only to score-inspired proposal adaptations, not reproductions of those methods. Thus this frozen layout is not evidence that simulations ran. Primary metrics are bias, RMSE, 95% coverage, interval width, false decisive actions, equivalence decisions, abstention, realized loss, maximum weight, and ESS, with Monte Carlo uncertainty. The simulation separately reports whether Bayes-risk acquisition actually lowers realized loss under the terminal confidence-set action.

The planned public study freezes all 200 roots at the pinned LiveBench reasoning revision, 200 hash-ranked roots from math, and all 128 coding roots: a 528-root weighted public-benchmark frame. Each source receives mass $1/3$ and roots are uniform within source. This is neither a product-domain sample nor a deployment population. It uses two validated prompt forms per item and paired response draws from two fixed open model revisions. Four model-judge families score all cells in both orders. The roots induce 2,112 root–prompt–order cells, 8,448 jury calls, and 168,960 human target units after crossing the 80-rater panel; human labels sample this frame rather than fully crossing it. A 200-rating target-random human pilot precedes 800 sequential ratings; a separate 200-rating target-random audit sampled from Equation (3) remains sealed. All counts and fixed budget $T = 1000$ assignment attempts are set by simulation before the first human outcome; a missing response is not replaced. A frozen panel of 80 raters is recruited against a written definition, assigned target weights, trained on a fixed handbook, and screened with non-study examples. The expected load is 12.5 primary attempts per rater; realized load, withdrawals, and response are reported. All T assignment slots are preassigned. Rater unavailability, withdrawal, or a binding quota records $R_t = 0$ in its original slot; no replacement occurs, later slots continue, and the analysis remains at fixed T using the predeclared partial-identification bounds. Identifiers are hidden, both display orders are balanced in expectation, and every skip or nonresponse is retained. Attempt position is randomized, sealed audit attempts are interleaved at hidden prespecified slots, and the drift sensitivity grid is frozen before labels.

6.1. Planned Primary Endpoints

Table 1. Primary finite-frame decision results. Dashes indicate that human collection has not been run. The table will report tie rate and interval, not a leaderboard rank.

Design	n_H	$\widehat{\Delta}_F$ [95% CI]	Tie	Action
Adaptive EVI	–	–	–	pending
Sealed audit	–	–	–	pending

Simulation will assess efficiency against valid common-cost baselines. The sealed audit is only a prospective concordance check; agreement cannot establish efficiency or validate either estimator. No separate human target-random, static-calibration, or PPAT arm is budgeted, so those rows are not reported as if they had been collected. Jury-only rankings are diagnostic and never substitute for Equation (2).

6.2. Robustness and Ablations

We ablate augmentation, EVI, the exploration floor, tie modeling, root-item effects, judge-family effects, and prequential fitting while crossing each acquisition policy with a common estimator. An exploratory extension would train acquisition on older model pairs and test a preregistered held-out stronger pair to expose frontier shift. Additional falsification tests swap AB/BA order, control response length and style, hold out prompt families, remove one judge family, change the practical margin, and repeat the study on a fresh temporal wave.

Model checks use held-out log score, multiclass Brier score, tie calibration, swap symmetry, posterior predictive checks, and residuals by domain, order, length, family, judge, and item. Disagreement cases are sampled by a prespecified stratified rule rather than selected for narrative convenience. SCOPE-style bidirectional conformal abstention is included as a jury-only diagnostic and never substitutes for the human estimand.

7. Limitations

The finite frame may omit important deployment requests, and a convenience rater panel supports inference only to that panel. Human preference is not factual correctness, safety, fairness, or welfare. The Davidson model can be misspecified, judge correlations can be more complex than modeled, and non-transitivity can make broader rankings incoherent. Sequential unbiasedness does not guarantee useful finite-sample intervals when exploration probabilities are small.

The method requires exact logging of every adaptive probability and a fixed budget or sequentially valid interval. EVI may save no labels and can be worse than target-random sampling under model shift. A sealed audit consumes additional human effort. Fresh items reduce exposure but impair longitudinal comparability, while black-box contamination cannot be ruled out. Finally, one pairwise replacement decision does not establish a general benchmark or model ordering.

8. Conclusion

We framed language-model comparison as a decision about a frozen, weighted human panel rather than a latent jury score. Randomized exploration and draw-level sequential augmentation permit outcome-adaptive human acquisition without surrendering design identification under explicit assumptions. The formal result does not show that acquisition is efficient or that the working model is valid. The decisive next evidence is coverage under misspecified, correlated simulations followed by prospective concordance with a sealed target-random human audit.

Impact Statement

Evaluation methods shape which systems are deployed and which capabilities are optimized. A narrow or unrepresentative item and rater frame can legitimize a harmful deployment even when its confidence interval is statistically valid. Adaptive labeling may undersample rare communities unless target weights and the exploration floor preserve them. Raters may encounter disturbing content; they require informed consent, task-specific warnings, the ability to skip without penalty, fair compensation, and access to support. Deployment logs must be de-identified and retained under strict access controls. Public reporting will include the population definition, tie rate, abstentions, maximum sampling weights, subgroup support, and unresolved validity threats. The method is not intended to turn human preference into a universal quality score or to justify deployment without separate safety and factuality audits.

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A. Datasheets and Reproducibility Details

A.1. Required Evaluation Manifest

The artifact records the frame timestamp, root-item source and target weight, deduplication cluster, domain and language, every prompt variant and weight, response coupling and seed, model and tokenizer revisions, decoder parameters, judge model and prompt revisions, order randomization, rater eligibility and training, rubric version, tie subtype, annotation cost, all $p_t(x)$ values before outcome reveal, outcome-model revision, EVI values, stopping rule, and exclusion reason. Probability normalization and positivity are checked online; a failed check halts collection.

The repository runner `python3 scripts/run_experiment_pipeline.py --plan experiments/eval_pipeline.json` enumerates frame acquisition, deduplication and response generation, jury-model construction, simulation and adaptive acquisition, and sealed evaluation. The committed JSON is a manifest validator and stage contract tied to the hashed evaluation section of `experiments/pilot_preregistration.json`; site-local commands are required for execution, after which output hashes and costs are recorded. The public-source registry pins each pilot frame and model revision. Automated license metadata is only a preflight; acquisition remains blocked until manual provenance, terms, privacy, human-subjects, and leakage reviews are approved. After those approvals, `scripts/materialize_public_data.py` verifies hash-pinned local exports, writes row-level provenance, and rejects retained content overlap across declared roles. Its only clearance bypass is a committed synthetic collision fixture whose artifacts are explicitly non-empirical. `scripts/run_content_gates.py` binds the frozen `experiments/content_gates.json` contract to those hashes and blocks retained named PII/secret patterns and lexical near-duplicates. Its large-artifact MinHash-LSH candidate generation can miss qualifying pairs, so that path is inconclusive by itself. `scripts/validate_external_overlap_audit.py` orders records by identifier, indexes the row-major upper triangle, and requires contiguous shards covering exactly $n(n-1)/2$ comparisons without gaps or overlaps. It binds every shard to the materialization, gate configuration, reviewed exact engine, runtime image, and review record; approximation, early termination, shard errors, or any qualifying pair blocks readiness. Reported pair identities, roles, thresholds, and scores are independently checked without storing matched text. A temporary disk-backed exact join additionally enumerates every pair sharing a 5-gram and rejects omissions; this is recall-complete for the positive frozen Jaccard thresholds because any qualifying pair must share a gram. Shard coverage is an engine-governance attestation, whereas readiness depends on reproducing the complete qualifying-pair set. The committed engine fields remain null blockers, so no large real frame can yet pass this route. An exact lexical pass still cannot establish semantic independence or approve a pending source review. The external contract indexes record pairs rather than canonical field segments, so it cannot clear a multi-field frame even after its engine fields are filled; a reviewed successor must cover every cross-record segment pair. `experiments/overlap_projection_contract.json` separately requires a source/revision/role inventory for every model-visible or protected field, including prompts, references, tests, and rubrics. Its validator binds canonical slots to source revision, configuration, split, exporter code, and a review-artifact hash. Runtime enumeration requires every textual leaf to be projected or explicitly excluded, and rejects a generated attestation relabeled as real. The generated fixture is a schema test only; all LiveBench field inventories remain null blockers. The same section fixes the Qwen, SmolLM, Mistral, and Phi jury revisions, posterior prior scales, one-step damped Newton approximation, unit annotation cost, nonresponse rule, and planned simulation factor levels. The separate `experiments/eval_simulation_design.json` replaces an infeasible full crossing with a deterministic 189-condition fractional design. All condition rows and seeds are stored explicitly in `experiments/eval_simulation_conditions.json`, not regenerated implicitly at execution. The validator reconstructs all 16 sign rows, rejects main–two-factor aliasing, checks every factor level against the preregistration, and derives condition seeds from canonical condition hashes. Generated fault injections for an aliased generator, stale condition count, and Cartesian-crossing flag must all fail. This is a design audit with `empirical_evidence=false`; executable DGPs, missingness laws, and acquisition-policy implementations remain blockers. These values specify a public pre-pilot; they do not replace a deployment-log sampling design. The adaptive acquisition contract also samples only a 256-unit target-random candidate multiset per batch before scoring EVI, retaining multiplicities in r_b. This preserves $p_b \geq \epsilon\pi$ while replacing an infeasible full-frame EVI pass with a frozen resource ceiling. Exceeding the ceiling blocks collection for that batch. The prompt-generator and rendered model prompts, variant weights, decoder and response-coupling seeds, response artifacts, and executable simulation DGPs are still unfrozen. Real response generation or human acquisition is blocked until a replacement preregistration hashes those estimand-defining objects. The companion `synthetic_commands.json` fixes a 16-unit frame, $\epsilon = 0.2$, $T = 300$, 2,000 trials, and seed 20260902. It uses deliberately misspecified prequential predictions to check the exact one-draw identity, positivity and weight bounds, fixed-budget studentization, and the tie estimator. It assumes complete response and is only a software audit; it does not replace the

preregistered condition grid or provide human evidence. The shared materialization validator now closes acquisition rows against retained, suppressed, and quarantined dispositions. `scripts/validate_eval_sequential_bundle.py` is the shared real/golden validator for the sequential decision bundle and frame hierarchy. It recomputes per-source SHA-256(revision‖question_id) root ranks, the selected roots and their weights w_i, the prompt-variant weights $a_{c|i}$, the root–prompt–order cells, the model-jury call count, and every human target unit's product mass $\pi(i, c, h, o) = w_i a_{c|i} \rho_h / 2$. From each batch commitment it recomputes the candidate-mixture law Equation (24), using log-sum-exp with log multiplicities. It rejects commitment-hash mismatches and fit slots that are not strictly earlier. It requires every one of the T fixed slots exactly once, so a missing outcome is recorded in place rather than replaced. From these inputs it reproduces Equation (8), the worst-case endpoints, the complete-outcome or martingale–Hoeffding interval, the tie interval, the drift widenings, the terminal action, the maximum weight, and the ESS. Its golden bundles set each EVI to $\log(a_x)/\lambda_E$ with integer a_x and use exact rational root, variant, order, and rater weights, so every draw probability and unit weight is an exact rational. The float recomputation must then match an independent `Fraction` implementation, and the self-test also reproduces the 528-root, 2,112-cell, 8,448-jury-call, 168,960-unit pilot hierarchy. Seventeen mutations fail, including a tampered root selection hash, a wrong hash-rank root selected, unnormalized variant weights, an incomplete cell crossing, a unit mass that disagrees with $w_i a_{c|i} \rho_h / 2$, a tampered p_t, EVI edited after commitment, a predictor fit on its own batch, a dropped or replacement slot, an outcome on a missing slot, and a complete-case interval reported despite missingness. A real bundle is refused while any estimand-defining artifact hash is null, its configuration differs from the preregistration, or it fails to bind `process/cell_manifest.json`. `scripts/recompute_eval_evi.py` now recomputes every committed EVI value under both the fixed-discrimination linear Davidson model ($\eta_x = f_x^\top \vartheta, \kappa_x = g_x^\top \vartheta$) and the jointly estimated log-discrimination bilinear Davidson model ($\eta_x = f_x^\top \vartheta + \exp(u_x^\top \vartheta)(s_x^\top \vartheta), \kappa_x = g_x^\top \vartheta$). Each adaptive batch's posterior artifact must carry the batch commitment hash and a frame design covering exactly the bundle frame; in the bilinear model it also enforces the human-panel identifiability constraint $\sum_x \pi(x) u_x = 0$. The validator then closes the posterior against the observed outcomes in that batch's strictly earlier fit slots. Using the exact parameter Jacobian $\nabla_\vartheta \eta_x = f_x + d_x s_x + (d_x q_x) u_x$ and observed negative Hessian $J_x^\top I_{(\eta,\kappa)} J_x - g_\eta(y) \nabla^2_{\vartheta\vartheta} \eta_x(\vartheta)$, the mode must be stationary for the Gaussian-prior Davidson log posterior, and the precision must equal prior precision plus observed information at the mode. Next it redraws the 4,096 common standard normals with seed 2026090322. It recomputes Q_b and the posterior predictive, applies the registered damped Newton update with ridge 10^{-4}, damping 10^{-3}, and the fixed jitter ladder for each hypothetical outcome, and requires every committed EVI to reproduce within 10^{-9}. Any draw within 10^{-9} of $\pm\delta$ fails closed. On both the three-parameter linear and four-parameter bilinear golden bundles with 79 observed fit outcomes, a pure-Python implementation that shares only the normal draws agrees with the numpy recomputation to 10^{-12}. Thirteen mutations fail: EVI set to zero, EVI assigned to the wrong unit, EVI from unregistered draws, a posterior fitted on its own batch, a precision without the prior, a stale commitment, a missing frame unit, an unregistered prior scale, a linear design mislabeled as bilinear, an unknown working model, a bilinear Jacobian that omits the $\log d_x$ chain-rule term, a bilinear Hessian that uses the Gauss–Newton term only, and an uncentered human-panel log-discrimination. The same self-test reports the EVI range, the count of negative values, and the maximum softmax tilt after 20, 40, 80, 120, and 200 generated fit outcomes; Section 4 quotes these design diagnostics. In a real train stage the runner now executes both the bundle validator and this EVI recomputation. The unfrozen prompt generator, paired-response coupling, executable simulation DGPs, and shared-rater adaptive-minus-audit covariance keep the artifact-integrity gate closed even if source reviews later pass.

A.2. Proof of Proposition 5.1

Conditioning on pre-outcome history,

$$\mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}] = \sum_x p_t(x) \left[\bar{m}_{t-1} + \frac{\pi(x)}{p_t(x)} \{\mu_x - m_{t-1}(x)\} \right] \tag{34}$$

$$= \bar{m}_{t-1} + \sum_x \pi(x) \{\mu_x - m_{t-1}(x)\} = \Delta_F. \tag{35}$$

Averaging over fixed T proves the second statement. Neither step requires a correct jury model.

A.3. Consistency of the Variance Estimate

Let $D_{Tt} = \psi_{Tt} - \Delta_F$. Fixed $\epsilon > 0$ makes D_{Tt} uniformly bounded. Hence $D_{Tt}^2 - \mathbb{E}[D_{Tt}^2 \mid \mathcal{F}_{t-1}]$ is a bounded martingale difference and its average converges to zero in probability. The conditional variance assumption in Theorem 5.2 then yields $T^{-1} \sum_t D_{Tt}^2 \to_p V$; replacing Δ_F by $\widehat{\Delta}_T$ changes the centered second moment by $o_p(1)$. Therefore $\widehat{V}_T \to_p V$ and

Slutsky's theorem justifies the stated studentized interval.

A.4. Human Audit

The audit draws complete (i, c, h, o) units from π before adaptation and is stored behind a separate random seed. It estimates the same finite-frame target from 200 fixed audit slots by the unweighted sample mean. Missing audit outcomes are replaced by -1 and $+1$ for lower and upper endpoint means, with one-sided Hoeffding radius $\sqrt{2\log(2/\alpha)/200}$; they are never dropped or replaced. If operational assignment changes the draw law, the exact target-to-draw ratio is retained and the corresponding bounded-weight interval is used. Audit membership never removes a unit from adaptive support: only the audit seed, assignments, and outcomes are hidden. Assignment seeds are independent, but persistent unobserved shared-rater session effects would violate Equation (6) unless included in the predictable history or conditioned upon. A clustered shared-rater contrast is therefore an assumption-violating sensitivity analysis, not a repair of the primary martingale claim. Its outcomes are opened only after the adaptive algorithm, model hyperparameters, and primary analysis are frozen. An adaptive-minus-audit equivalence test remains blocked until its margin, dependence model, and failure rule are frozen.

B. Additional Validity Evidence

Table 2. Mandatory validity report; all values are pending collection.

Diagnostic	Value
Simulation coverage at $\Delta_F = \delta$	–
False decisive-action rate	–
Maximum $\pi(X_t)/p_t(X_t)$	–
Adaptive-sample ESS	–
Human inter-rater agreement	–
Jury multiclass Brier score	–
AB/BA swap discrepancy	–
Adaptive–audit estimate difference	–
Temporal-wave drift	–

Full per-domain estimates, tie subtypes, disagreement matrices, acquisition trajectories, all simulation conditions, and randomly selected qualitative cases will be reported. Post-hoc analyses are labeled exploratory and cannot replace the endpoints in Table 1.